

# MATH 350-2 Advanced Calculus

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# Outline

- 1 Real Analysis Lecture 7
  - More on component intervals
  - Closed Sets

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- $(-\infty, 0)$  is also a component interval of  $\mathbb{R} \setminus \{0, 1, 2, 3\}$
- we will show all open sets of  $\mathbb{R}$  are made of component intervals!

# Component intervals

## Theorem (Apostol 3.10)

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Claim:  $(a, b)$  is a component interval of  $U$  containing  $x$ .  $\square$

# Challenge

## Problem

Prove that  $(a, b)$  is a component interval of  $U$ .

# Open subsets of $\mathbb{R}$

## Theorem (Representation Theorem for Open Sets in $\mathbb{R}$ )

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Uh oh ... this isn't a **countable** union.



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Since  $x$  was arbitrary,

$$U = \bigcup_{r \in U \cap \mathbb{Q}} I_r$$



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## Examples:

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- products of closed intervals

# Challenge!

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Prove that a singleton set

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Since  $\vec{x}$  was arbitrary, this proves  $U$  is open.

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Prove that the **closed square**

$$[a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}$$

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Union of open sets is open!

# Adherent and Accumulation Points

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- suprema and infima are accumulation points!
- $0$  is an accumulation point of  $\{1/1, 1/2, 1/3, \dots\}$

# Characterizing accumulation points

## Theorem (Apostol Theorem 3.17)

*A point  $\vec{x}$  is an accumulation point of  $A$  if and only if for all  $r > 0$ , the ball  $B(\vec{x}; r)$  contains infinitely many points of  $A$ .*

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Suppose instead that  $\vec{x}$  is an accumulation point and let  $r > 0$ . It suffices to show that the set  $C = (B(\vec{x}; r) \setminus \{\vec{x}\}) \cap A$  is infinite.



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We conclude that  $C$  is infinite. □

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## Theorem (Apostol Theorem 3.18, 3.20, 3.22)

*A set is closed if and only if it contains all of its adherent points (ie.  $A = \bar{A}$ ), or equivalently if and only if  $A$  contains all of its accumulation points.*

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# Closures are closed

## Theorem

*If  $A \subseteq \mathbb{R}^n$  is any set, then  $\overline{A}$  is closed.*

## Proof.

Exercise! □

# Closed balls are closed

## Problem

Show that the closed ball

$$\overline{B}(\vec{x}; r) = \{\vec{y} \in \mathbb{R}^n : |\vec{x} - \vec{y}| \leq r\}$$

is a closed set.